MATH 231 – Linear Algebra I

Lecture 1 ▪ Vectors: Operations and the Geometry of $\mathbb{R}^2$

Tuesday, September 1, 2026 ▪ Kiely Hall 326 ▪ Instructor: Kevin Bedoya

What you should be able to do afterwards. State what a vector is in terms of magnitude and direction; identify the tail and head of a vector in the plane; compute the Euclidean norm and explain why it is a *unary* operation; distinguish vector quantities from scalar quantities; add and subtract vectors both componentwise and geometrically, using the parallelogram rule and the triangle inequality; scale a vector by a real number and say what $\alpha = -1$ does to it; decide when two vectors drawn in different places are the *same* vector; say what it means for two vectors to be parallel or orthogonal, in terms of the angle θ between them; derive $\langle \mathbf{u}, \mathbf{v} \rangle = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta$ from the law of cosines; compute dot products and cosine similarities; test two vectors for orthogonality with a single dot product; normalize a vector and split it into a direction and a magnitude; compute the projection of one vector onto another and the component that goes with it; split a vector into a piece along $\mathbf{u}$ and a piece orthogonal to $\mathbf{u}$; and state the Cauchy–Schwarz inequality.

Further reading. Boyd & Vandenberghe, *Introduction to Applied Linear Algebra*, Ch. 1 (vectors, addition, scalar multiplication) and Ch. 3 (norm, distance, angle); Strang, *Linear Algebra and Its Applications*, Ch. 1; Axler, *Linear Algebra Done Right*, Ch. 1; Trefethen & Bau, *Numerical Linear Algebra*, Lec. 2–3 (norms, orthogonality).

1. Vectors: magnitude, direction, and the space they live in

We begin with a working definition — deliberately informal, and stated in the language of quantities rather than axioms.

Definition 1 (Vector, working definition)**.** A *vector* is a mathematical object carrying two attributes: a **magnitude** and a **direction**.

Looking ahead. This is the physicist's definition, and it is the right one to start from because it is the one you can *see*. Later in the course we replace it with the algebraist's definition: a vector will become any element of a set equipped with two operations obeying a short list of axioms — at which point functions, polynomials, and matrices all become vectors too, and "magnitude and direction" stops being the defining feature. For now, hold on to the picture.

The space a vector lives in

A vector does not exist in isolation; it lives in a space. Ours today is the **two-dimensional Euclidean plane**, the familiar xy-plane, and every vector we draw will live in it.

The word *dimension* here means a **degree of freedom** with respect to spatial coordinates: how many independent numbers must you supply to pin down a location? On a line, one; on the plane, two — an x and a y, chosen independently of one another. The plane is two-dimensional in exactly that sense.

Tail, head, and the two attributes

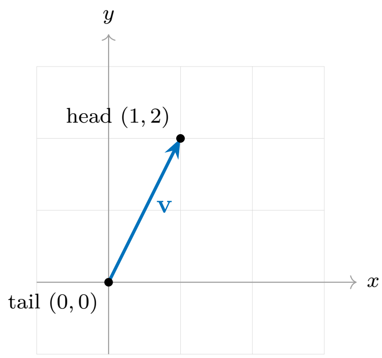
Fix the point $(1, 2)$. Draw an arrow from the origin to it. That arrow is a vector, and it has two distinguished points:

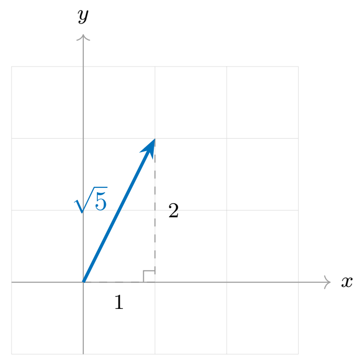
- the **tail** — where the arrow begins, here the origin $(0, 0)$;
- the **head** — where the arrow ends and the arrowhead is drawn, here $(1, 2)$.

With the tail and head named, the two attributes become concrete:

- **Direction** is the arrow's orientation in the plane — which way it points, relative to the axes, as it travels from tail to head. The arrow to $(1, 2)$ points up and to the right, more steeply up than rightward.
- **Magnitude** is the arrow's **length**: the distance from the tail to the head. We measure it with the Euclidean distance, and we compute it in §2.

Note the arrowhead in the left-hand figure below: the arrowhead *is* the direction, so a bare line segment would lose half the information.





2. The norm — our first vector operation

The magnitude of a vector is not just a description of it; it is something we *compute* from it. Computing it is our first vector operation, and it has a name.

Definition 2 (Euclidean norm)**.** For a vector $\mathbf{v}$ in the plane with tail at the origin and head at (x, y), the **Euclidean norm** of $\mathbf{v}$ is

$$\|\mathbf{v}\| \;=\; \sqrt{x^2 + y^2},$$

the Euclidean distance from the tail to the head.

Two things to notice about the shape of this operation.

- It is **unary**: it takes *one* vector as input. Compare $3+5$, where addition consumes two numbers — a *binary* operation. The norm consumes one vector and returns one number.
- It turns a *vector* into a *scalar*: the input is a vector in the plane, the output is a single real number, and that number is never negative. Length cannot be negative.

The formula is Pythagoras, nothing more. The right-hand figure above is the proof: the horizontal leg has length x, the vertical leg has length y, and the vector is the hypotenuse.

Example 1 (The running vector)**.** For $\mathbf{v}$ with head at $(1, 2)$,

$$\|\mathbf{v}\| \;=\; \sqrt{1^2 + 2^2} \;=\; \sqrt{1+4} \;=\; \sqrt{5} \;\approx\; 2.236.$$

Note that $\sqrt{5}$ is a finished answer: a norm is generically irrational, and there is no need to convert it to a decimal.

Example 2 (A clean one)**.** For $\mathbf{w}$ with head at $(3, 4)$, $\|\mathbf{w}\| = \sqrt{9+16} = \sqrt{25} = 5$.

Remark. So three phrases now name the same number: the *magnitude* of the vector, the *length* of the arrow, and the *Euclidean norm* $\|\mathbf{v}\|$. Use them interchangeably.

3. Bridge to calculus: the derivative as a vector quantity

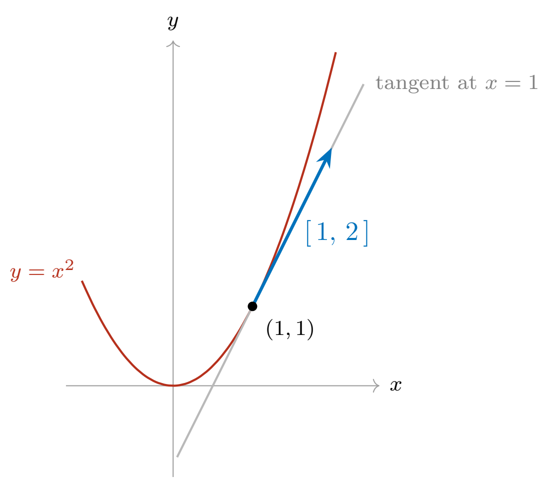
Vectors are not new to you — you have been computing one all through calculus without calling it that.

Let f be differentiable and consider the tangent line to the curve $y = f(x)$ at a point. The tangent line has both of our attributes:

- **Direction.** The *sign* of the slope orients it. A positive slope tilts up-and-to-the-right and the function is increasing there; a negative slope tilts down-and-to-the-right and the function is decreasing. The tangent line points somewhere.
- **Magnitude.** The *size* of the slope measures how fast the function is changing — the "speed" of change in that direction. A slope of 10 climbs ten times as urgently as a slope of 1.

Direction plus magnitude: the derivative of f at a point can be read as a **vector quantity**.

Concretely, take $f(x) = x^2$ at $x = 1$. Then $f'(1) = 2$, so from the point $(1, 1)$ the tangent advances 1 unit rightward for every 2 units upward: its direction vector is $[\,1,\ 2\,]$ — the very vector we drew in §1, now carrying a meaning. Its norm, $\sqrt{5}$, is the rate at which arc length accumulates along the tangent per unit of horizontal travel.



4. Vector quantities vs. scalar quantities

The phrase from §3 is worth making official, because it sorts most of the quantities you will meet in the sciences into two bins.

- A **vector quantity** has a magnitude *and* a direction. *Examples:* displacement, velocity, acceleration, force, momentum, torque, the electric and magnetic fields, a temperature *gradient.*
- A **scalar quantity** has a magnitude only — it indicates *scale*, and nothing else. *Examples:* mass, speed, temperature, energy, work, electric charge, time, pressure, density, volume.

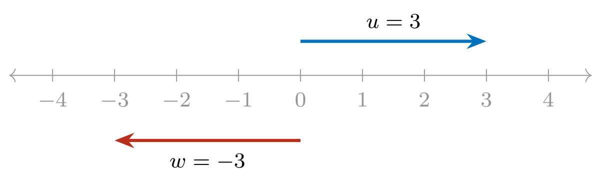
The pair to dwell on is **speed** and **velocity**. Speed is a scalar: 60 mph. Velocity is a vector: 60 mph *due north*. Speed is exactly the magnitude of the velocity vector — that is, its norm — which is why the two words are not synonyms and why a car going round a bend at constant speed is nonetheless accelerating.

5. One-dimensional vectors

Nothing above required two dimensions. In one dimension — on the number line — a vector is a signed number, and the two attributes are still both present:

- the **sign** carries the direction: positive points one way along the line, negative the other;
- the **absolute value** carries the magnitude.

So $u = 3$ and $w = -3$ have the same magnitude and opposite directions. And the one-dimensional norm is $\|u\| = \sqrt{u^2} = |u|$ — absolute value was your first norm all along.



6. Notation, and where vectors sit set-theoretically

There are several standard ways to denote a vector symbolically. Using the letter v purely as an example — *any* letter may be used:

Notation	**Reading**
$\mathbf{v}$	boldface — common in applied and computational writing
$\vec{v}$	arrow overhead — common in physics and calculus texts
v	plain — common in pure linear algebra, where context suffices

In this class we will use a mixture of all three, and context will tell you which object you are looking at. Getting comfortable reading all three is part of the course: you will meet all of them in the reference texts.

The set-theoretic statement

Until now we have said "the plane" in words. Here is its name. To say that $\mathbf{v}$ is a two-dimensional vector, we write

$$\mathbf{v} \in \mathbb{R}^2, \qquad \text{where} \qquad \mathbb{R}^2 \;=\; \mathbb{R} \times \mathbb{R} \;=\; \big\{(x, y) \;:\; x \in \mathbb{R},\; y \in \mathbb{R}\big\}.$$

Read aloud: "$\mathbf{v}$ is in $\mathbb{R}$-two." The superscript 2 is the dimension of §1 — two coordinates, each free to be any real number, chosen independently. From here on, $\mathbb{R}^2$ replaces "the plane" as our standard name for the space a two-dimensional vector lives in.

The numerical convention for this class

We write a vector in brackets, listing its coordinates in order:

$$\mathbf{v} = [\,x,\,y\,], \qquad \text{for instance} \qquad \mathbf{v} = [\,1,\,2\,].$$

Order matters: $[1,2] \neq [2,1]$.

Looking ahead. For now we draw every vector with its tail at the origin, so that its head sits at (x, y). That is a convention, not a restriction: in §9 we will see that a vector is really a *displacement*, so the same vector may be drawn with its tail anywhere in the plane. That fact is what makes the head-to-tail picture of §7.1 legitimate.

7. Addition and subtraction

Many operations on vectors await us. Here are the first two.

Definition 3 (Addition and subtraction in $\mathbb{R}^2$)**.** Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ with $\mathbf{u} = [x_1, y_1]$ and $\mathbf{v} = [x_2, y_2]$. Then

$$\mathbf{u} + \mathbf{v} = [\,x_1 + x_2,\;\; y_1 + y_2\,], \qquad \mathbf{u} - \mathbf{v} = [\,x_1 - x_2,\;\; y_1 - y_2\,].$$

Both are **componentwise**: the first coordinates interact only with each other, and likewise the second. Nothing mixes across slots.

Now notice the *shape* of these two operations, exactly as we did for the norm in §2. Each one is a **binary operator**: it consumes *two* vectors and returns a vector.

$$+ \;:\; \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^2, \qquad\qquad - \;:\; \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^2.$$

Read the left-hand side as "a pair of vectors from $\mathbb{R}^2$ goes in." Contrast this with the norm, which was *unary* — one vector in, one *number* out. Addition and subtraction are binary, and they never leave $\mathbb{R}^2$: two vectors in, one vector out.

Example 3 (Both operations on one pair)**.** Let $\mathbf{u} = [2, 1]$ and $\mathbf{v} = [1, 3]$. Then

$$\mathbf{u} + \mathbf{v} = [\,2 + 1,\; 1 + 3\,] = [\,3,\; 4\,], \qquad \mathbf{u} - \mathbf{v} = [\,2 - 1,\; 1 - 3\,] = [\,1,\; -2\,].$$

And since we know how to take a norm: $\|\mathbf{u} + \mathbf{v}\| = \sqrt{9 + 16} = 5$ — the clean vector from §2, arrived at by addition.

7.1. The parallelogram rule

Componentwise arithmetic tells you *what* the sum is. Geometry tells you *where* it is.

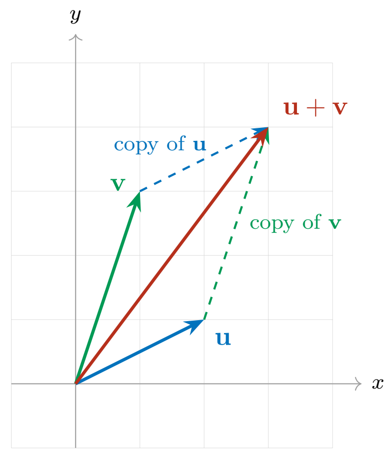
> **Parallelogram rule.** Place $\mathbf{u}$ and $\mathbf{v}$ tail-to-tail at the origin. They span a parallelogram: the two given arrows are one pair of sides, and the copies of them drawn from the opposite corners are the other pair. Then $\mathbf{u}+\mathbf{v}$ is the **diagonal of that parallelogram** running from the common tail to the far corner.

There is a second reading of the same picture, and it is the one to carry around in your head:

> **Head to tail.** Walk along $\mathbf{u}$. From where you land, walk along $\mathbf{v}$. The sum $\mathbf{u} + \mathbf{v}$ is the single arrow from where you started to where you finished.

The two readings agree because the far corner of the parallelogram is reached either way. Notice what the head-to-tail reading quietly does: it draws $\mathbf{v}$ with its tail at the head of $\mathbf{u}$ rather than at the origin. That is legitimate — it is the same vector $\mathbf{v}$, just drawn elsewhere — and §9 is where we make that precise.

In the figure below, the solid arrows are $\mathbf{u}$, $\mathbf{v}$ and their sum from the origin; the dashed arrows are the translated copies that close the parallelogram. Follow the blue solid arrow and then the green dashed one: you arrive at $[3,4]$, which is the head of the red diagonal.



7.2. The triangle inequality

Take the head-to-tail picture and ignore the parallelogram: what is left is a *triangle*. Its three sides have lengths $\|\mathbf{u}\|$, $\|\mathbf{v}\|$ and $\|\mathbf{u}+\mathbf{v}\|$. And a triangle can never have one side longer than the other two combined — walking straight from start to finish cannot be longer than detouring via a third point.

Triangle inequality. For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$,

$$\|\mathbf{u}+\mathbf{v}\| \;\leq\; \|\mathbf{u}\| + \|\mathbf{v}\|.$$

Equality holds in exactly one situation: when $\mathbf{u}$ and $\mathbf{v}$ point in the *same* direction (or one of them is the zero vector). Then the "triangle" has collapsed flat onto a single line, the detour is no detour at all, and the two sides lie end to end along the third.

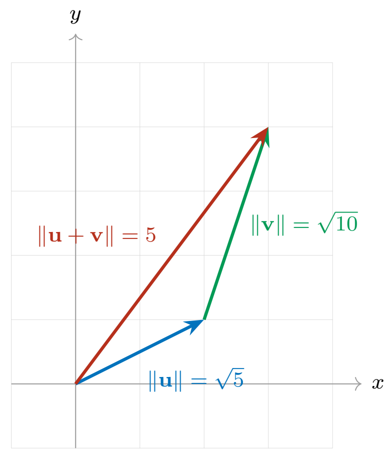
Example 4 (Strict inequality)**.** With $\mathbf{u} = [2,1]$ and $\mathbf{v} = [1,3]$ from above, $\mathbf{u}+\mathbf{v} = [3,4]$ and

$$\|\mathbf{u}+\mathbf{v}\| = 5, \qquad \|\mathbf{u}\| + \|\mathbf{v}\| = \sqrt{5} + \sqrt{10} \approx 2.236 + 3.162 = 5.398.$$

Indeed $5 < 5.398$. The two vectors point in genuinely different directions, so the detour costs something.

Example 5 (Equality)**.** Let $\mathbf{u} = [1,2]$ and $\mathbf{v} = [2,4]$. Here $\mathbf{v}$ is just $\mathbf{u}$ made twice as long — same direction — so we expect equality. And $\mathbf{u}+\mathbf{v} = [3,6]$, so

$$\|\mathbf{u}+\mathbf{v}\| = \sqrt{9+36} = \sqrt{45} = 3\sqrt{5}, \qquad \|\mathbf{u}\| + \|\mathbf{v}\| = \sqrt{5} + 2\sqrt{5} = 3\sqrt{5}.$$



This inequality is not a curiosity. It is one of the two or three properties that *define* what it means to be a norm at all, and it is the reason distances behave the way our intuition expects.

7.3. Subtraction: the arrow between the heads

Subtraction has its own picture, and it needs no new machinery.

> **Between the heads.** Draw $\mathbf{u}$ and $\mathbf{v}$ from the origin. Then $\mathbf{u} - \mathbf{v}$ is the arrow joining the two heads, pointing *from* the head of $\mathbf{v}$ *to* the head of $\mathbf{u}$.

It is the displacement that carries you from $\mathbf{v}$ to $\mathbf{u}$ — which is why differences of vectors are how we will measure how far apart two vectors are.

That leaves one thing to get right, and it is the step students most often reverse: *which way does the arrowhead go?* Here is a rule that never fails.

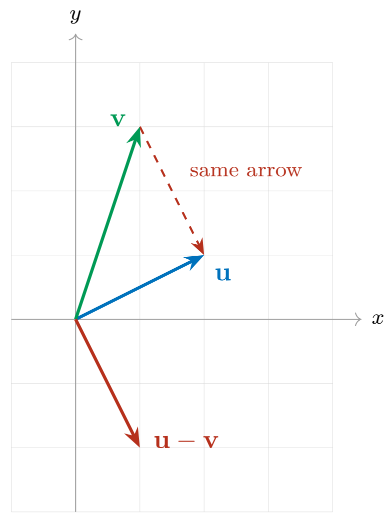
> **Reading a difference as a journey.** Read $\mathbf{v} - \mathbf{u}$ from right to left: *start at* $\mathbf{u}$, *end at* $\mathbf{v}$. So $\mathbf{v} - \mathbf{u}$ points *at* $\mathbf{v}$, and $\mathbf{u} - \mathbf{v}$ points *at* $\mathbf{u}$. The arrowhead always lands on the head of whichever vector is written *first*.

And a numerical check, in case the picture ever betrays you: compute the components and read off their signs. For $\mathbf{v} - \mathbf{u}$, the first coordinate tells you whether the arrow travels right (positive) or left (negative), and the second tells you up (positive) or down (negative).

Example 6 (Both directions on one pair)**.** Take $\mathbf{u} = [2, 1]$ and $\mathbf{v} = [1, 3]$ again.

$$\mathbf{u} - \mathbf{v} = [\,1,\ -2\,], \qquad \mathbf{v} - \mathbf{u} = [\,-1,\ 2\,].$$

By the journey rule, $\mathbf{u} - \mathbf{v}$ starts at the head of $\mathbf{v}$, namely $(1, 3)$, and ends at the head of $\mathbf{u}$, namely $(2, 1)$ — one step right and two steps down, matching $[\,1, -2\,]$. The other difference, $\mathbf{v} - \mathbf{u} = [-1, 2]$, is the same segment travelled backwards: one step left and two steps up. The two are opposite arrows on the same line.



The dashed arrow and the solid arrow labelled $\mathbf{u} - \mathbf{v}$ are the *same vector* drawn in two places: same magnitude, same direction, different tail. §9 is where that claim gets its definition.

8. Scalar multiplication

Addition and subtraction combine a vector with another vector. The next operation combines a vector with a *scalar* — one of the plain numbers of §4, carrying magnitude but no direction.

Definition 4 (Scalar multiplication in $\mathbb{R}^2$)**.** Let $\mathbf{v} = [\,x,\ y\,] \in \mathbb{R}^2$ and let $\alpha \in \mathbb{R}$. Then

$$\alpha\mathbf{v} \;=\; [\,\alpha x,\;\; \alpha y\,].$$

Componentwise again: α multiplies *every* coordinate.

This operation is binary as well, but its two inputs are of *different kinds* — a number and a vector, not two vectors:

$$\mathbb{R} \times \mathbb{R}^2 \to \mathbb{R}^2.$$

A scalar and a vector go in; a vector comes out. That mixed signature is why α is called a *scalar* in the first place: its job is to set the scale of the vector.

Example 7 (Scaling the running vector)**.** Let $\mathbf{v} = [\,1,\ 2\,]$, our vector from §1, with $\|\mathbf{v}\| = \sqrt{5}$.

$$2\mathbf{v} = [\,2,\ 4\,], \qquad \tfrac{1}{2}\mathbf{v} = [\,\tfrac{1}{2},\ 1\,], \qquad -\mathbf{v} = (-1)\mathbf{v} = [\,-1,\ -2\,].$$

Their norms are

$$\|2\mathbf{v}\| = \sqrt{4+16} = \sqrt{20} = 2\sqrt{5}, \qquad \|\tfrac{1}{2}\mathbf{v}\| = \sqrt{\tfrac{1}{4}+1} = \tfrac{\sqrt{5}}{2}, \qquad \|-\mathbf{v}\| = \sqrt{1+4} = \sqrt{5}.$$

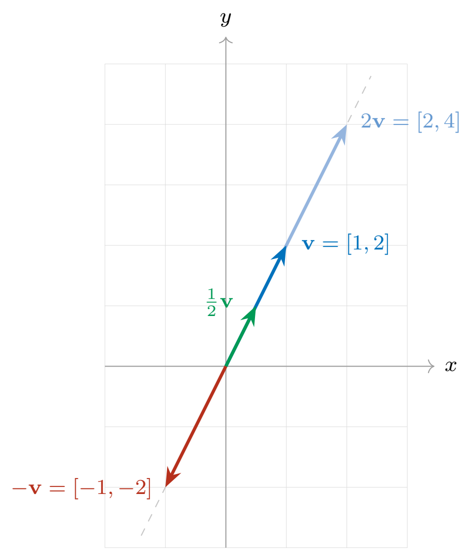
So $2\mathbf{v}$ is twice as long as $\mathbf{v}$, $\frac{1}{2}\mathbf{v}$ is half as long, and $-\mathbf{v}$ is exactly as long as $\mathbf{v}$ but points the opposite way.

Remark. Read off the pattern: scaling by α multiplies the magnitude by $|\alpha|$,

$$\|\alpha\mathbf{v}\| = |\alpha|\,\|\mathbf{v}\|,$$

and the *sign* of α decides the direction — positive keeps it, negative reverses it. That is the same sign-carries-direction rule you saw on the number line in §5, now acting on a vector in the plane. Taking $\alpha = 0$ collapses the vector to $[\,0,\ 0\,]$, the **zero vector**, which has no direction at all.

Geometrically, scalar multiplication never turns a vector off its own line: every multiple of $\mathbf{v}$ lies on the single straight line through the origin and the head of $\mathbf{v}$. Scaling slides the arrowhead along that line — stretching it out, pulling it in, or flipping it through the origin — and nothing else.



8.1. What scaling does to the picture

The algebra is one line; the geometry is worth tabulating, because the *sign* and the *size* of α do two independent jobs. The size $|\alpha|$ decides the length; the sign decides the direction.

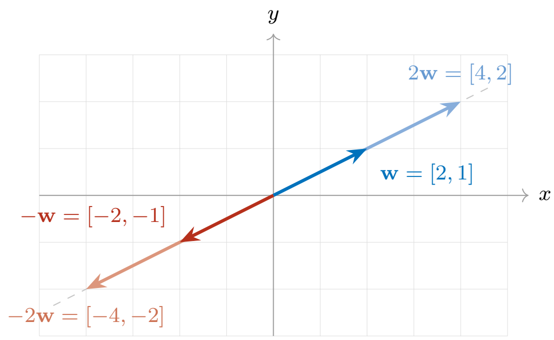
Scalar	Effect on length	Effect on direction
$\alpha > 1$	stretched	unchanged
$\alpha = 1$	unchanged	unchanged
$0 < \alpha < 1$	shrunk	unchanged
$\alpha = 0$	collapses to the zero vector	destroyed
$-1 < \alpha < 0$	shrunk	reversed
$\alpha = -1$	unchanged	reversed
$\alpha < -1$	stretched	reversed

The figure above showed three of these rows on $\mathbf{v} = [1, 2]$: $\alpha = 2$ stretched it, $\alpha = \frac{1}{2}$ shrank it, $\alpha = -1$ reversed it. The row still missing a picture is the last one — reversing *and* stretching at once — so here it is on a flatter vector, to make the point that none of this is special to $[1, 2]$.

Example 8 (Reversing and stretching together)**.** Let $\mathbf{w} = [\,2,\ 1\,]$, so $\|\mathbf{w}\| = \sqrt{5}$. Then

$$2\mathbf{w} = [\,4,\ 2\,], \qquad -\mathbf{w} = [\,-2,\ -1\,], \qquad -2\mathbf{w} = [\,-4,\ -2\,].$$

Read the last one geometrically: $\alpha = -2$ is negative, so the arrow turns around; and $|\alpha| = 2$, so it comes out twice as long. All four arrows lie on one line through the origin — the two positive multiples on one side, the two negative multiples on the other.



The table also runs backwards, which is often how you will use it: if you can *see* two arrows on a common line through the origin, you can read off the scalar that relates them without computing anything.

Example 9 (Reading the scalar off a picture)**.** An arrow is drawn on the same line through the origin as $\mathbf{v} = [1, 2]$. It points the *opposite* way, and it is *three times as long*. Which multiple of $\mathbf{v}$ is it?

Opposite direction forces $\alpha < 0$; three times as long forces $|\alpha| = 3$. Together, $\alpha = -3$, so the arrow is

$$-3\mathbf{v} = [\,-3,\ -6\,].$$

Checking against the algebra: $\|-3\mathbf{v}\| = \sqrt{9+36} = \sqrt{45} = 3\sqrt{5} = 3\|\mathbf{v}\|$, as the picture promised.

Example 10 (Scaling a physical vector)**.** Recall from §4 that velocity is a vector quantity. Let $\mathbf{v}$ be the velocity of a car travelling northeast at 30 mph. Then:

- $2\mathbf{v}$ is northeast at 60 mph — same heading, twice the speed;
- $\frac{1}{2}\mathbf{v}$ is northeast at 15 mph — same heading, half the speed;
- $-\mathbf{v}$ is southwest at 30 mph — same speed, reversed heading;
- $0\mathbf{v}$ is the car at rest, with no heading at all.

Scaling changes the speed and, if the scalar is negative, the heading — but it never turns the car onto a new bearing that is neither the original nor its exact reverse.

8.2. Multiplication by -1: mirroring

The case $\alpha = -1$ deserves a name of its own, because it is the one you will use constantly.

> **Mirroring.** $-\mathbf{v} = (-1)\mathbf{v}$ is the **mirror image of v through the origin**: same line, same length, opposite direction. Every coordinate flips sign.

In the figure above, $\mathbf{v} = [1, 2]$ and $-\mathbf{v} = [-1, -2]$ sit on opposite sides of the origin along the same dashed line, at equal distances from it. The origin is the midpoint of the segment joining their two heads.

Remark. Be precise about which mirror. Reflecting *through the origin* — a point — flips *both* coordinates: $[1, 2] \mapsto [-1, -2]$. Reflecting *across an axis* — a line — flips only one: across the x-axis, $[1, 2] \mapsto [1, -2]$. Multiplication by -1 is the first of these, not the second. Equivalently, it is a half-turn: rotation by π about the origin.

8.3. Subtraction revisited: $\mathbf{u} + (-\mathbf{v})$

In §7.3 we read $\mathbf{u} - \mathbf{v}$ off the picture as the arrow between the two heads. Now that we own the mirror, a second reading opens up — and it lets us reuse the parallelogram rule instead of learning a separate recipe.

Subtraction is nothing but addition of the mirror image:

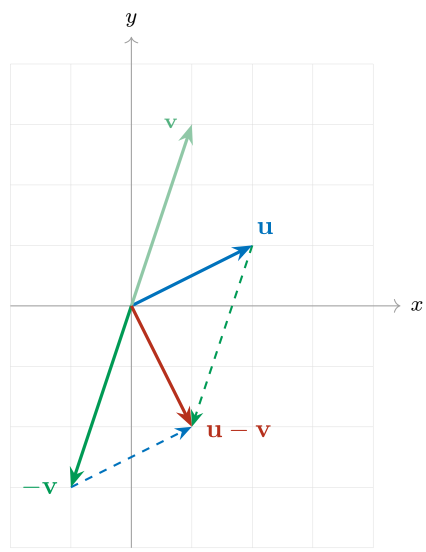
$$\mathbf{u} - \mathbf{v} \;=\; \mathbf{u} + (-\mathbf{v}).$$

So to subtract geometrically: **mirror $\mathbf{v}$ through the origin to get $-\mathbf{v}$, then add $\mathbf{u}$ and $-\mathbf{v}$ by the parallelogram rule of §7.1.** No new geometry at all — one reflection, then the rule you already have.

Example 11 (The same difference, computed two ways)**.** Take $\mathbf{u} = [2, 1]$ and $\mathbf{v} = [1, 3]$ once more. Mirroring gives $-\mathbf{v} = [-1, -3]$, and the parallelogram rule on $\mathbf{u}$ and $-\mathbf{v}$ produces the diagonal

$$\mathbf{u} + (-\mathbf{v}) = [\,2 + (-1),\;\; 1 + (-3)\,] = [\,1,\, -2\,],$$

which is exactly the $\mathbf{u} - \mathbf{v}$ we found in §7.3. Two different pictures, one vector.



The faint green arrow is the original $\mathbf{v}$; the solid green one is its mirror image $-\mathbf{v}$, and the faint dashed line through the origin shows that the two really do lie on a common line. The red diagonal of the parallelogram spanned by $\mathbf{u}$ and $-\mathbf{v}$ is $\mathbf{u} - \mathbf{v}$.

9. When are two vectors equal?

We have twice now drawn "the same vector" in two different places and asked you to believe it. Time to define it.

Definition 5 (Equality of vectors)**.** Two vectors are **equal** when they have the same magnitude *and* the same direction. In components, for $\mathbf{u} = [u_1, u_2]$ and $\mathbf{v} = [v_1, v_2]$,

$$\mathbf{u} = \mathbf{v} \qquad \Longleftrightarrow \qquad u_1 = v_1 \;\text{ and }\; u_2 = v_2.$$

Notice what the definition does *not* mention: where the arrow is. A vector is a magnitude together with a direction — a *displacement* — and a displacement does not care where you start it.

Components from any tail: head minus tail

So how do you read the components off an arrow that does not start at the origin? Subtract.

> An arrow with tail $A = (a_1, a_2)$ and head $B = (b_1, b_2)$ is the vector
> $$[\, b_1 - a_1,\;\; b_2 - a_2 \,] \qquad \text{—} \qquad \textbf{head minus tail}.$$

This is not a new rule; it is the subtraction of §7.3 seen from the other side. Two arrows are the same vector precisely when their head-minus-tail differences agree, no matter how far apart the arrows are drawn. Every vector therefore has infinitely many drawings, called *representatives*, and exactly one of them has its tail at the origin — that one is said to be in **standard position**, and it is the drawing we have been defaulting to since §1.

Example 12 (Three drawings, one vector)**.** Each of these arrows is the vector $[\,2,\ 1\,]$:

(i) tail $(0, 0)$, head $(2, 1)$: $\quad [\,2 - 0,\ 1 - 0\,] = [\,2,\ 1\,]$;

(ii) tail $(1, 3)$, head $(3, 4)$: $\quad [\,3 - 1,\ 4 - 3\,] = [\,2,\ 1\,]$;

(iii) tail $(-3, -1)$, head $(-1, 0)$: $\quad [\,-1 - (-3),\ 0 - (-1)\,] = [\,2,\ 1\,]$.

They sit in three different places on the page. They are the same vector: each one says “two steps right, one step up.”

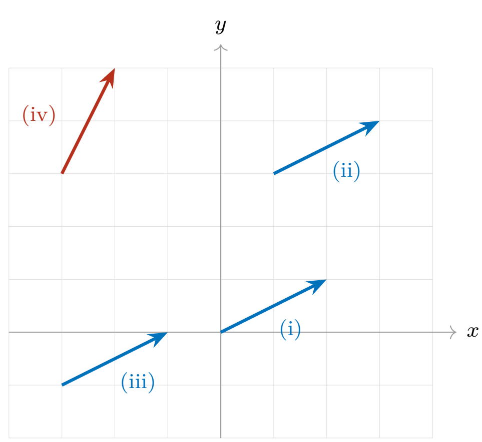
Example 13 (Same magnitude is not enough)**.** $[\,1,\ 2\,]$ and $[\,2,\ 1\,]$ both have norm $\sqrt{5}$, so they are equally long — but the first climbs steeply and the second runs flat. Different directions, so they are *different vectors*.

Example 14 (Same direction is not enough either)**.** $[\,1,\ 2\,]$ and $[\,2,\ 4\,]$ point exactly the same way — the second is 2 times the first, so by §8 they lie on one line through the origin — but their norms are $\sqrt{5}$ and $2\sqrt{5}$. Different magnitudes, so again *different vectors*.

Example 15 (Mirrors are not equal)**.** $[\,1,\ 2\,]$ and $[\,-1,\ -2\,]$ have the same norm $\sqrt{5}$ and lie on the same line, but they point opposite ways. By §8.2 the second is the mirror image of the first, and a mirror image is a *different vector* — unless the vector is $[\,0,\ 0\,]$, which is its own mirror.

Equality needs *both* attributes to match. The last three examples fail on exactly one apiece.

A common confusion. An arrow drawn near the top of the page and an arrow drawn near the bottom *look* like two different objects, and it is tempting to treat them as two different vectors. They are not, so long as head minus tail comes out the same for both. Conversely, two arrows can start at the very same point and still be different vectors. Position on the page is not part of the data; magnitude and direction are.



Arrows (i), (ii) and (iii) are the three representatives of $[\,2,\,1\,]$ from the example above: parallel, equally long, all pointing the same way, and therefore *one and the same vector*. Arrow (iv) is $[\,1,\,2\,]$; it has the same length $\sqrt{5}$ as the others but a different direction, so it is a different vector.

10. Relative orientation: parallel, orthogonal, and the angle between

Two vectors sitting in $\mathbb{R}^2$ have a relationship beyond their individual magnitudes and directions: how they are oriented *with respect to each other*. Two arrangements are special enough to be named.

10.1. Parallel vectors

> **Parallel (geometric definition).** Two nonzero vectors are **parallel** when, drawn from a common tail, they lie along one and the same straight line. Equivalently: the lines containing their arrows are parallel lines, so the arrows never tilt away from each other — they either point the same way or exactly opposite ways.

§8 already drew this picture without naming it: every scalar multiple of $\mathbf{v}$ lands on the single line through the origin and the head of $\mathbf{v}$. That is precisely the algebraic version of the definition.

> **Parallel (algebraic test).** Nonzero $\mathbf{u}$ and $\mathbf{v}$ are parallel exactly when one is a scalar multiple of the other: $\mathbf{v} = \alpha\mathbf{u}$ for some $\alpha \neq 0$. If $\alpha > 0$ they point the *same* way; if $\alpha < 0$ they point *opposite* ways, and the pair is often called **antiparallel**.

Example 16 (Parallel, both signs)**.** $[\,1,\,2\,]$ and $[\,3,\,6\,]$ are parallel and point the same way, since $[3,6] = 3[1,2]$ with $\alpha = 3 > 0$.
$[\,1,\,2\,]$ and $[\,-2,\,-4\,]$ are parallel and antiparallel in direction, since $[-2,-4] = (-2)[1,2]$ with $\alpha = -2 < 0$.
$[\,1,\,2\,]$ and $[\,2,\,1\,]$ are *not* parallel: no single α satisfies both $2 = \alpha \cdot 1$ and $1 = \alpha \cdot 2$.

The zero vector is excluded from the definition on purpose: it has no direction, so asking whether it is parallel to anything is asking a question with no content.

10.2. Orthogonal vectors

> **Orthogonal.** Two nonzero vectors are **orthogonal** when, drawn from a common tail, they meet at a right angle. "Orthogonal" is the word we will use throughout the course; "perpendicular" means the same thing.

The cleanest example is the pair of coordinate directions themselves: $[\,1,\,0\,]$ and $[\,0,\,1\,]$ meet at a right angle at the origin. A less obvious pair is $[\,2,\,1\,]$ and $[\,-1,\,2\,]$: each is the other turned a quarter turn, and both have norm $\sqrt{5}$.

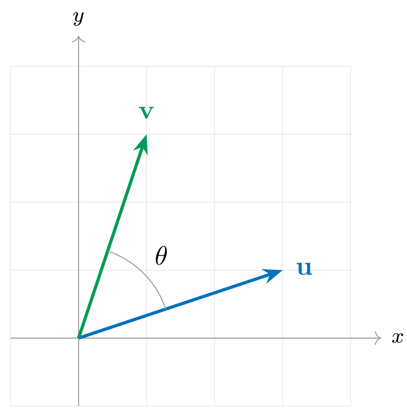
Remark. You already own one test for that last claim: from precalculus, two lines are perpendicular when their slopes multiply to -1. The arrows $[2,1]$ and $[-1,2]$ have slopes $\frac{1}{2}$ and -2, and $\frac{1}{2}\cdot(-2) = -1$. But notice how brittle this test is. It says nothing at all when one of the vectors is vertical — slope undefined — it does not extend to three dimensions or beyond, and it answers only the yes/no question. It cannot tell you how far from perpendicular two vectors are when they are not.

10.3. The angle between two vectors

"Parallel" and "orthogonal" are the two extreme cases of a single underlying quantity, and everything in between has been left unnamed. To measure relative orientation *numerically*, in a way that covers all cases at once, we need the **angle between the vectors**.

Definition 6 (The angle θ)**.** Let $\mathbf{u}$ and $\mathbf{v}$ be nonzero vectors in $\mathbb{R}^2$, drawn from a common tail. Write θ for the angle they open up at that shared tail, measured so that

$$\theta \in [\,0,\ \pi\,].$$

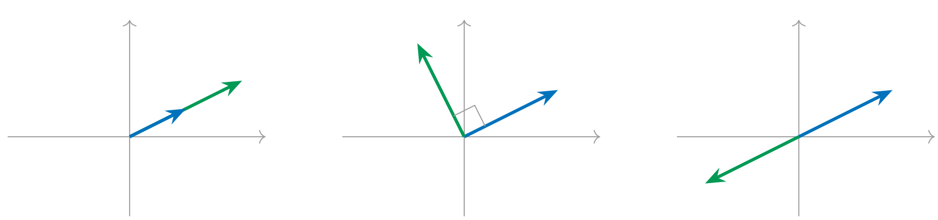


Why that range? Two rays leaving a common point cut the plane into two angles, and those two angles add up to 2π. One of them is the opening *between* the rays; the other is the reflex angle going the long way around outside. Restricting θ to $[0, \pi]$ always selects the first one. Two consequences, both of which we want:

- θ is genuinely the angle *between* the two vectors, never the one sweeping around behind them;
- θ does not depend on which vector you start from — the angle from $\mathbf{u}$ to $\mathbf{v}$ equals the angle from $\mathbf{v}$ to $\mathbf{u}$. It is a property of the *pair*, with no orientation or sign attached.

With θ in hand, both named arrangements become single numbers, and the unnamed middle ground gets a scale:

Angle	**Arrangement**	**Name**
$\theta = 0$	same direction	parallel
$0 < \theta < \frac{\pi}{2}$	opening less than a right angle	acute
$\theta = \frac{\pi}{2}$	right angle	**orthogonal**
$\frac{\pi}{2} < \theta < \pi$	opening more than a right angle	obtuse
$\theta = \pi$	opposite directions	antiparallel



$\theta = 0$: parallel $\qquad$ $\theta = \pi/2$: orthogonal $\qquad$ $\theta = \pi$: antiparallel

Looking ahead. We can now *say* what relative orientation means, and we could measure θ off the page with a protractor. What we cannot yet do is *compute* it from the components $[u_1, u_2]$ and

$[v_1, v_2]$ — nothing we have built so far turns four numbers into an angle. The tool that does it is the *inner product*, and §11 constructs it. It will collapse the whole table above into arithmetic: in particular, testing two vectors for orthogonality will become a single multiplication-and-addition, with no protractor, no slopes, and no exceptions for vertical vectors.

11. The dot product and cosine similarity

Section 10 left us in an awkward position. We can *say* precisely what parallel, antiparallel and orthogonal mean — they are $\theta = 0$, $\theta = \pi$ and $\theta = \pi/2$ — but θ is not something the world hands us. Vectors arrive as pairs of numbers, not as angles, and no amount of staring at $[u_1, u_2]$ and $[v_1, v_2]$ produces a protractor. So:

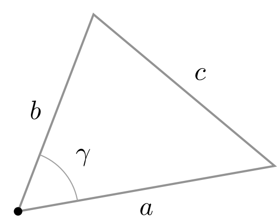
> Given only the components of two vectors, how do we decide *numerically* whether they are parallel, antiparallel, or orthogonal?

The answer comes from a theorem you already know, applied to a triangle we already know how to build.

11.1. Review: the law of cosines

The law of cosines generalises Pythagoras to triangles that have no right angle.

> **Law of cosines.** In a triangle with two sides of lengths a and b meeting at an angle γ, the third side — the one *opposite* γ — has length c given by
>
> $$c^2 \;=\; a^2 + b^2 - 2ab\cos\gamma.$$



Taking $\gamma = \pi/2$ gives $\cos\gamma = 0$ and the formula collapses to $c^2 = a^2 + b^2$ — Pythagoras is the right-angled special case. The extra term $-2ab\cos\gamma$ is exactly the correction for the angle not being right.

11.2. The triangle built from $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{u} - \mathbf{v}$

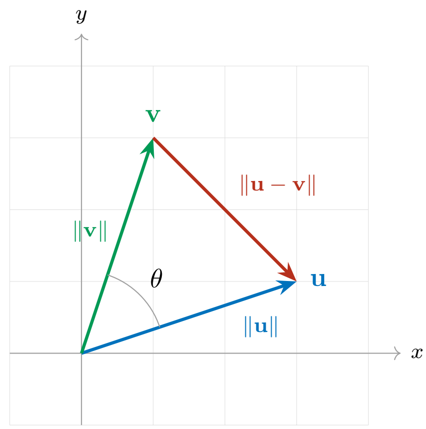
Now build a triangle out of two vectors. Draw $\mathbf{u}$ and $\mathbf{v}$ from a common tail at the origin. Their two heads, together with that shared tail, are three points — a triangle. Two of its sides are the vectors themselves. And the third side, the one joining the two heads, is something we identified back in §7.3: it is $\mathbf{u} - \mathbf{v}$.

So the triangle has side lengths

$$a = \|\mathbf{u}\|, \qquad b = \|\mathbf{v}\|, \qquad c = \|\mathbf{u} - \mathbf{v}\|,$$

and the angle where sides a and b meet — at the common tail — is exactly θ, the angle between $\mathbf{u}$ and $\mathbf{v}$ from §10.3. The side of length c sits opposite it. That is precisely the configuration the law of cosines describes, with $\gamma = \theta$:

$$\|\mathbf{u} - \mathbf{v}\|^2 \;=\; \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta. \tag{1}$$



The derivation

Equation (1) is pure geometry: it contains θ but no components. Now compute the same left-hand side *algebraically*, from the components, and compare. With $\mathbf{u} = [u_1, u_2]$ and $\mathbf{v} = [v_1, v_2]$ we have $\mathbf{u} - \mathbf{v} = [\, u_1 - v_1,\ u_2 - v_2\,]$, so

$$\begin{aligned}
\|\mathbf{u}-\mathbf{v}\|^2 &= (u_1 - v_1)^2 + (u_2 - v_2)^2 \\
&= \big(u_1^2 - 2u_1v_1 + v_1^2\big) + \big(u_2^2 - 2u_2v_2 + v_2^2\big) \\
&= \underbrace{\big(u_1^2 + u_2^2\big)}_{\|\mathbf{u}\|^2} + \underbrace{\big(v_1^2 + v_2^2\big)}_{\|\mathbf{v}\|^2} - 2\big(u_1v_1 + u_2v_2\big) \\
&= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\big(u_1v_1 + u_2v_2\big).
\end{aligned}$$

Set this equal to the geometric expression (1):

$$\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\big(u_1v_1 + u_2v_2\big) \;=\; \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta.$$

The squared norms cancel from both sides, and dividing by -2 leaves the identity we were after:

$$\boxed{u_1v_1 + u_2v_2 \;=\; \|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta} \tag{2}$$

where θ is the angle between $\mathbf{u}$ *and* $\mathbf{v}$. That last clause is not decoration: the identity is false for any other angle, and every use we make of (2) depends on θ being the angle between the two vectors in question.

Look at what (2) accomplishes. The left side is built from *components alone* — two multiplications and one addition, no geometry. The right side contains θ. The equation is a bridge between the two, and it is the bridge §10.3 said we were missing.

11.3. The dot product

The left-hand side of (2) is important enough to name.

Definition 7 (Dot product / inner product in $\mathbb{R}^2$)**.** For $\mathbf{u} = [u_1, u_2]$ and $\mathbf{v} = [v_1, v_2]$ in $\mathbb{R}^2$, the **dot product** — equivalently the **inner product** — of $\mathbf{u}$ and $\mathbf{v}$ is

$$\langle \mathbf{u}, \mathbf{v} \rangle \;=\; \mathbf{u} \cdot \mathbf{v} \;=\; u_1v_1 + u_2v_2.$$

Both notations are standard and we will use both: $\mathbf{u} \cdot \mathbf{v}$ is common in applied writing, $\langle \mathbf{u}, \mathbf{v} \rangle$ in pure linear algebra. With the name in place, (2) becomes the statement we will quote constantly:

$$\langle \mathbf{u}, \mathbf{v} \rangle \;=\; \|\mathbf{u}\|\,\|\mathbf{v}\| \cos\theta, \qquad \theta = \text{the angle between } \mathbf{u} \text{ and } \mathbf{v}.$$

Remark (A notation trap). Do *not* write $\mathbf{u} \times \mathbf{v}$ for this. That symbol denotes the **cross product**, a completely different operation: it is defined in $\mathbb{R}^3$ rather than $\mathbb{R}^2$, and it returns a *vector* rather than a scalar. Writing $\times$ when you mean $\cdot$ is one of the most common notational errors in a first linear algebra course. Use $\cdot$ or $\langle \cdot\,, \cdot \rangle$, and reserve $\times$ for the cross product.

Where it sits among our operations

The dot product is our newest **binary operator** — two vectors in, one result out. But notice what makes it structurally new: it is the first binary operator that *leaves* $\mathbb{R}^2$.

$$\langle \cdot\,, \cdot \rangle \;:\; \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}.$$

Two vectors go in and a *scalar* comes out. Addition and subtraction were binary and stayed inside $\mathbb{R}^2$; the norm left $\mathbb{R}^2$ but was unary. The dot product is the first operation to do both at once.

Operation	**Signature**	**Arity**	**Returns**
Norm $\|\cdot\|$ (§2)	$\mathbb{R}^2 \to \mathbb{R}$	unary	scalar
Addition $+$ (§7)	$\mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^2$	binary	vector
Subtraction $-$ (§7)	$\mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^2$	binary	vector
Scalar multiplication (§8)	$\mathbb{R} \times \mathbb{R}^2 \to \mathbb{R}^2$	binary (mixed)	vector
Dot product $\langle \cdot\,, \cdot \rangle$ (§11)	$\mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$	binary	scalar

11.4. Cosine similarity

Solve (2) for the cosine. Since $\theta \in [0, \pi]$ by §10.3, and cosine is decreasing from $\cos 0 = 1$ to $\cos\pi = -1$ across that interval, we know in advance that

$$\cos\theta \in [-1, 1].$$

Dividing (2) by $\|\mathbf{u}\|\|\mathbf{v}\|$ — legal whenever neither vector is the zero vector — gives

$$\frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|\,\|\mathbf{v}\|} \;=\; \cos\theta.$$

The left-hand side is computable from components. So we may as well promote it to a function in its own right.

Definition 8 (Cosine similarity)**.** For nonzero $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$, define

$$\cos(\mathbf{u}, \mathbf{v}) \;:=\; \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|\,\|\mathbf{v}\|}.$$

This is a function taking *two* vectors in $\mathbb{R}^2$ and returning a real number in $[-1, 1]$. It is called the **cosine similarity** of $\mathbf{u}$ and $\mathbf{v}$, and by the identity above it equals $\cos\theta$, where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$.

The name earns itself twice over. "Cosine," because the output *is* the cosine of the angle between the two vectors. "Similarity," because that output is a score: the closer the two vectors point in the same direction, the closer it is to 1.

Remark (Order does not matter). Cosine similarity is **symmetric**:

$$\cos(\mathbf{u}, \mathbf{v}) = \cos(\mathbf{v}, \mathbf{u}).$$

Both the dot product and the product of norms are unchanged when the two vectors swap places, and §10.3 already established that θ is a property of the *pair* with no preferred starting vector.

A word on the terminology, since the adjectives are easy to swap: a two-argument function is *symmetric* when the *order* of its inputs does not matter, which is what the display above says. *Reflexive* is a different property — it concerns what happens when both inputs are the *same* object. Cosine similarity has a clean property of that kind too, and we compute it in §11.6.

Remark (Scale does not matter either). Because $\cos(\mathbf{u}, \mathbf{v})$ depends only on θ, stretching either vector by a positive factor leaves it unchanged: $\cos(\alpha\mathbf{u}, \mathbf{v}) = \cos(\mathbf{u}, \mathbf{v})$ for every $\alpha > 0$. Cosine similarity measures *direction agreement only* and is blind to magnitude — which is exactly what you want from a similarity score, and exactly what distinguishes it from the dot product itself.

Remark (Similarity kernels). In computational mathematics a function that takes two objects and returns a scalar score of how alike they are is called a **similarity kernel**. Cosine similarity is the first one we meet, and it is by far the most widely used in practice. The objects need not be vectors in $\mathbb{R}^2$ — kernels are built for text documents, images, graphs, and molecules — and we will meet other kernels later in the course.

Reading the three special values

Cosine similarity answers the question this section opened with. Each of the three named arrangements corresponds to one exact value, and the correspondence is made by inverting the cosine on $[0, \pi]$, where arccos is defined and single-valued:

- $\cos(\mathbf{u}, \mathbf{v}) = 1$. Then $\theta = \arccos(1) = 0$: the two vectors open up no angle at all, so they point the same way. $\mathbf{u}$ and $\mathbf{v}$ are **parallel**.
- $\cos(\mathbf{u}, \mathbf{v}) = 0$. Then $\theta = \arccos(0) = \pi/2$: the two vectors form a right angle. $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal**.
- $\cos(\mathbf{u}, \mathbf{v}) = -1$. Then $\theta = \arccos(-1) = \pi$: the opening is a straight line, so the vectors point in exactly opposite directions. $\mathbf{u}$ and $\mathbf{v}$ are **antiparallel**.

Compare this against the table in §10.3. Every row of that table, which we could then only describe, is now a number we can compute from four components.

11.5. When are two vectors numerically similar?

The three special values are the exact cases. What makes cosine similarity useful in practice is everything in between: two vectors are said to be **numerically similar** when $\cos(\mathbf{u}, \mathbf{v}) \approx 1$ — close to pointing the same way, without being exactly parallel.

Example 17 (A ladder of similarity)**.** Take $\mathbf{u} = [\,1,\ 0\,]$, so $\|\mathbf{u}\| = 1$, and compare it against several partners.

$\mathbf{v}$	$\langle \mathbf{u}, \mathbf{v} \rangle$	$\cos(\mathbf{u}, \mathbf{v})$	**Reading**
$[\,7,\, 1\,]$	7	$7/\sqrt{50} \approx 0.990$	nearly the same direction
$[\,3,\, 1\,]$	3	$3/\sqrt{10} \approx 0.949$	similar
$[\,1,\, 1\,]$	1	$1/\sqrt{2} \approx 0.707$	45° apart
$[\,0,\, 1\,]$	0	0	orthogonal — unrelated
$[\,-7,\, 1\,]$	-7	$-7/\sqrt{50} \approx -0.990$	nearly opposite

The first two rows are what "numerically similar" means. The fourth is the neutral case. The last is as dissimilar as two vectors get.

Example 18 (Similar but nowhere near equal)**.** $\mathbf{u} = [\,3,\, 4\,]$ and $\mathbf{v} = [\,300,\, 400\,]$ have $\langle \mathbf{u}, \mathbf{v} \rangle = 900 + 1600 = 2500$, $\|\mathbf{u}\| = 5$ and $\|\mathbf{v}\| = 500$, so

$$\cos(\mathbf{u}, \mathbf{v}) = \frac{2500}{5 \cdot 500} = 1.$$

They are perfectly similar — *parallel* — even though one is a hundred times longer than the other. This is the blindness to magnitude at work: by §9 these are emphatically *not* equal vectors, but their directions agree exactly, and direction is all cosine similarity looks at.

$\cos \approx 0.99$ $\cos \approx 0.71$ $\cos = 0$ $\cos \approx -0.99$

In each panel $\mathbf{u}$ is the horizontal arrow and $\mathbf{v}$ is the other one. Both are drawn at the same length on purpose: cosine similarity cannot see length, only the opening between them.

11.6. A vector against itself

Before computing anything, guess. What should $\cos(\mathbf{u}, \mathbf{u})$ be?

A vector is as similar to itself as anything could possibly be, and it opens no angle at all with itself, so $\theta = 0$ and we expect

$$\cos(\mathbf{u}, \mathbf{u}) \;=\; 1.$$

Now verify it algebraically, straight from the definition. For $\mathbf{u} = [u_1, u_2]$ nonzero,

$$\cos(\mathbf{u}, \mathbf{u}) = \frac{\langle \mathbf{u}, \mathbf{u} \rangle}{\|\mathbf{u}\|\,\|\mathbf{u}\|} = \frac{u_1 u_1 + u_2 u_2}{\|\mathbf{u}\|^2} = \frac{u_1^2 + u_2^2}{\|\mathbf{u}\|^2}.$$

And now look at that numerator. By the definition of the norm in §2, $\|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2}$, so $u_1^2 + u_2^2$ is exactly $\|\mathbf{u}\|^2$. The numerator and denominator are the same number:

$$\cos(\mathbf{u}, \mathbf{u}) = \frac{\|\mathbf{u}\|^2}{\|\mathbf{u}\|^2} = 1,$$

as intuition promised. But the observation we made along the way is worth more than the result:

Identity. For every $\mathbf{u} \in \mathbb{R}^2$,

$$\langle \mathbf{u}, \mathbf{u} \rangle \;=\; \|\mathbf{u}\|^2, \qquad \text{equivalently} \qquad \|\mathbf{u}\| \;=\; \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}.$$

Dotting a vector with itself returns its squared length. This will do a great deal of work for us in computations and proofs later in the course: it is what lets us convert a statement about lengths into a statement about dot products, and back again, whenever a proof needs one form rather than the other. It is also the first hint of something deeper — that the norm was never independent of the inner product, but a consequence of it.

11.7. The orthogonality test

One special case is used so often that it deserves to be separated from the rest.

Suppose $\mathbf{u}$ and $\mathbf{v}$ are nonzero and orthogonal. Then $\theta = \pi/2$, and $\cos(\pi/2) = 0$, so (2) reads

$$\langle \mathbf{u}, \mathbf{v} \rangle = \|\mathbf{u}\|\,\|\mathbf{v}\| \cos\!\left(\tfrac{\pi}{2}\right) = \|\mathbf{u}\|\,\|\mathbf{v}\| \cdot 0 = 0.$$

Conversely, if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ with both vectors nonzero, then $\|\mathbf{u}\|\|\mathbf{v}\| \cos\theta = 0$; the two norms are strictly positive, so $\cos\theta = 0$, and the only angle in $[0, \pi]$ with zero cosine is $\theta = \pi/2$. The implication runs both ways.

Orthogonality test. For nonzero $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$,

$$\mathbf{u} \text{ and } \mathbf{v} \text{ are orthogonal} \qquad \Longleftrightarrow \qquad \langle \mathbf{u}, \mathbf{v} \rangle = 0.$$

This is the payoff promised in §10.3. To test two vectors for orthogonality you do *not* need the angle, the cosine similarity, the norms, or a protractor — and none of the slope-test exceptions apply. Two multiplications, one addition, and check whether the answer is zero.

Example 19 (The test in action)**.** Are $[\,2,\,1\,]$ and $[\,-1,\,2\,]$ orthogonal? Compute $\langle [2,1], [-1,2] \rangle = 2(-1) + 1(2) = -2 + 2 = 0$. Yes. This confirms by arithmetic what §10.2 argued with slopes — and unlike the slope test, it would work just as well against the vertical vector $[\,0,\,1\,]$: $\langle [1,0], [0,1] \rangle = 0$.

11.8. Practice: computing cosine similarity

Let

$$\mathbf{u} = [\,1,\,2\,], \qquad \mathbf{v} = [\,2,\,4\,], \qquad \mathbf{w} = [\,-2,\,1\,], \qquad \mathbf{z} = [\,-3,\,-6\,].$$

Compute $\cos(\mathbf{u}, \mathbf{v})$, $\cos(\mathbf{u}, \mathbf{w})$ and $\cos(\mathbf{u}, \mathbf{z})$, and in each case name the arrangement. Then compute $\cos(\mathbf{v}, \mathbf{w})$.

The norms you will need: $\|\mathbf{u}\| = \sqrt{5}$, $\|\mathbf{v}\| = \sqrt{20} = 2\sqrt{5}$, $\|\mathbf{w}\| = \sqrt{5}$, $\|\mathbf{z}\| = \sqrt{45} = 3\sqrt{5}$.

Answer key

Pair	Dot product	Cosine similarity	Arrangement
$\mathbf{u}, \mathbf{v}$	$1(2) + 2(4) = 10$	$\dfrac{10}{\sqrt{5} \cdot 2\sqrt{5}} = \dfrac{10}{10} = 1$	parallel
$\mathbf{u}, \mathbf{w}$	$1(-2) + 2(1) = 0$	$\dfrac{0}{\sqrt{5} \cdot \sqrt{5}} = 0$	orthogonal
$\mathbf{u}, \mathbf{z}$	$1(-3) + 2(-6) = -15$	$\dfrac{-15}{\sqrt{5} \cdot 3\sqrt{5}} = \dfrac{-15}{15} = -1$	antiparallel
$\mathbf{v}, \mathbf{w}$	$2(-2) + 4(1) = 0$	$\dfrac{0}{2\sqrt{5} \cdot \sqrt{5}} = 0$	orthogonal

Remark (Why four vectors and not three?). Because three cannot exhibit all three arrangements. To realise "parallel" or "antiparallel" at all, two of the three vectors must lie on a common line through the origin — and once they do, a third vector's relation to one of them determines its relation to the other. That is exactly what the last row above illustrates: $\mathbf{v} = 2\mathbf{u}$ lies on $\mathbf{u}$'s line, so $\mathbf{w}$, being orthogonal to $\mathbf{u}$, has no choice but to be orthogonal to $\mathbf{v}$ as well. Two of the three pairs therefore always carry the same label, and {parallel, antiparallel, orthogonal} can never appear once each. A fourth vector is the minimum needed.

12. Unit vectors, projection, and Cauchy–Schwarz

Since §1 we have said that a vector carries two attributes, a magnitude and a direction. So far we have only ever handled them together. This section pulls them apart — a **direction–magnitude decomposition**, if you like — and then uses the pieces to answer a genuinely new question: how much of one vector points along another?

12.1. Normalization and unit vectors

To isolate the direction of a vector, throw away its magnitude by dividing it out.

Definition 9 (Normalization)**.** A nonzero vector $\mathbf{u} \in \mathbb{R}^2$ is **normalized** by dividing it by its own magnitude. We write

$$\hat{\mathbf{u}} \;:=\; \frac{\mathbf{u}}{\|\mathbf{u}\|} \;=\; \frac{1}{\|\mathbf{u}\|}\,\mathbf{u},$$

read "$\mathbf{u}$-hat." The result has length 1, and a vector of length 1 is called a **unit vector**; we also say it has **unit length**.

Why does normalizing produce length 1? Because dividing by $\|\mathbf{u}\|$ is scalar multiplication by $\alpha = 1/\|\mathbf{u}\|$, and §8 told us exactly what scaling does to a norm:

$$\|\hat{\mathbf{u}}\| = \left\| \tfrac{1}{\|\mathbf{u}\|}\,\mathbf{u} \right\| = \left| \tfrac{1}{\|\mathbf{u}\|} \right| \|\mathbf{u}\| = \frac{\|\mathbf{u}\|}{\|\mathbf{u}\|} = 1.$$

And since $1/\|\mathbf{u}\| > 0$, the sign rule of §8.1 says the direction is untouched. Normalizing therefore does precisely one thing: it discards the magnitude and keeps the direction. That is the whole point — we normalize when we want to talk about a vector's direction *only*.

Example 20 (Normalizing)**.** For $\mathbf{u} = [\,3,\,4\,]$ we have $\|\mathbf{u}\| = 5$, so

$$\hat{\mathbf{u}} = \frac{1}{5}[\,3,\,4\,] = [\,0.6,\,0.8\,],$$

and indeed $\|\hat{\mathbf{u}}\| = \sqrt{0.36 + 0.64} = \sqrt{1} = 1$. The vector $\hat{\mathbf{u}}$ points the same way as $\mathbf{u}$ but is five times shorter.

The zero vector cannot be normalized — the division is undefined, which is the arithmetic shadow of the fact recorded in §8 that $[\,0,\,0\,]$ has no direction to extract.

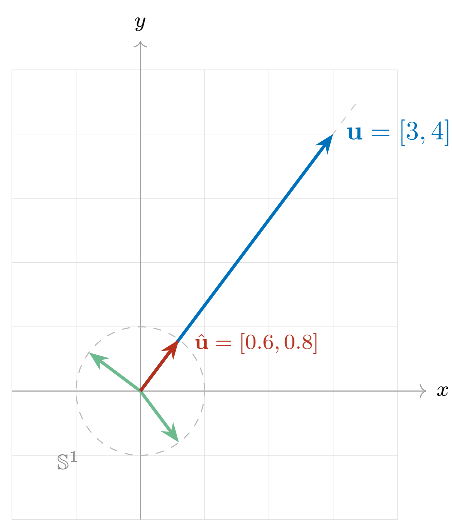
Where all the unit vectors live

A unit vector satisfies $\|\mathbf{u}\| = 1$, that is $\sqrt{u_1^2 + u_2^2} = 1$, or equivalently

$$u_1^2 + u_2^2 = 1.$$

That is the equation of a circle of radius 1 centred at the origin. So:

> Under the Euclidean norm, the heads of *all* unit vectors lie on a single circle of radius 1 about the origin — and every point of that circle is the head of a unit vector.



Looking ahead. The circle is an artefact of the *Euclidean* norm, not of normalization itself. Later in the course we will define other norms on $\mathbb{R}^2$, and each one has its own set of "vectors of length 1" — which turns out to be a different shape in the plane for each norm. The circle is only the first of a family of pictures.

The set-theoretic statement

That circle has a name, and it lets us say "$\mathbf{u}$ has unit length" in the notation of §6.

Definition 10 (The 1-sphere)**.**

$$\mathbb{S}^1 \;:=\; \big\{\, \mathbf{u} \in \mathbb{R}^2 \;:\; \|\mathbf{u}\| = 1 \,\big\}.$$

Writing $\mathbf{u} \in \mathbb{S}^1$ is a formal way of saying that $\mathbf{u}$ is a unit vector.

Remark. The superscript counts the dimension of the *sphere itself*, not of the space it sits in. The 1-sphere $\mathbb{S}^1$ is a circle: a one-dimensional curve living in the two-dimensional plane — you need only one number, an angle, to say where you are on it. The 2-sphere $\mathbb{S}^2$ is the ordinary sphere: a two-dimensional surface living in three-dimensional space. So a circle is a "one-dimensional sphere," and the object you would casually call "a sphere" is $\mathbb{S}^2$.

12.2. The direction–magnitude decomposition

Now put the two attributes back together deliberately. Multiply and divide any nonzero $\mathbf{u}$ by its own norm:

$$\mathbf{u} \;=\; \underbrace{\frac{\mathbf{u}}{\|\mathbf{u}\|}}_{\text{direction}} \cdot \underbrace{\|\mathbf{u}\|}_{\text{magnitude}} \;=\; \hat{\mathbf{u}}\,\|\mathbf{u}\|.$$

The first factor is a unit vector carrying the direction and nothing else; the second is a positive scalar carrying the magnitude and nothing else. Every nonzero vector splits this way, and the split is unique:

> **Direction–magnitude decomposition.** For every nonzero $\mathbf{u} \in \mathbb{R}^2$ there are a unit vector $\hat{\mathbf{u}} \in \mathbb{S}^1$ and a scalar $\alpha \in \mathbb{R}$ with $\alpha > 0$ such that
>
> $$\mathbf{u} \;=\; \alpha\,\hat{\mathbf{u}},$$
>
> and they are uniquely determined: $\alpha = \|\mathbf{u}\|$ and $\hat{\mathbf{u}} = \mathbf{u}/\|\mathbf{u}\|$.

Remark. Two fine points, both worth stating once. First, the restriction $\alpha > 0$ is what buys uniqueness: if negative scalars were allowed we could also write $\mathbf{u} = (-\|\mathbf{u}\|)(-\hat{\mathbf{u}})$, since $-\hat{\mathbf{u}}$ is a unit vector too by §8.2. Second, the zero vector is genuinely excluded — it has no direction, so there is no unit vector to factor out of it.

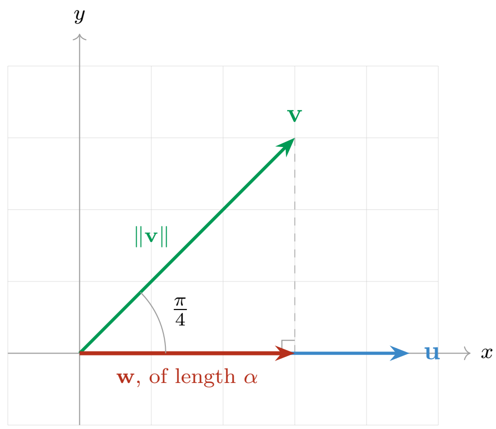
From here on, whenever we want "a vector pointing that way, with that length," we will build it as (scalar) $\times$ (unit vector). The next subsection is the first place we need to.

12.3. The shadow of one vector on another

Here is the question. Let $\mathbf{u}$ be a vector lying along the x-axis, and let $\mathbf{v}$ be a vector making an angle of 45° — that is, $\pi/4$ — with $\mathbf{u}$.

> How much of $\mathbf{v}$ points in the direction of $\mathbf{u}$?

Put physically: if light shone straight down onto the line carrying $\mathbf{u}$, what *shadow* would $\mathbf{v}$ cast on it? Call that shadow $\mathbf{w}$. To find it, drop a line from the head of $\mathbf{v}$ perpendicular to $\mathbf{u}$; where it lands is the head of the shadow.



Deriving the shadow

Step 1 — the direction. The shadow lies along $\mathbf{u}$, so it points in the direction $\hat{\mathbf{u}} = \mathbf{u}/\|\mathbf{u}\|$. By the decomposition of §12.2 we may therefore write it as

$$\mathbf{w} \;=\; \frac{\mathbf{u}}{\|\mathbf{u}\|} \cdot \alpha,$$

where α is the length of the shadow. This is exactly why we built the decomposition first: we know the direction, we do not yet know the magnitude, so we write the vector in the form that separates them.

Step 2 — the length, by trigonometry. The perpendicular we dropped creates a right triangle. Its hypotenuse is $\mathbf{v}$, of length $\|\mathbf{v}\|$; the leg along $\mathbf{u}$ is the shadow, of length α; and the angle between them is $\pi/4$. Cosine is adjacent over hypotenuse, so

$$\cos\frac{\pi}{4} = \frac{\alpha}{\|\mathbf{v}\|}, \qquad \text{hence} \qquad \alpha = \|\mathbf{v}\|\cos\frac{\pi}{4}.$$

Step 3 — replace the angle by the cosine similarity. Here $\pi/4$ is the angle between $\mathbf{u}$ and $\mathbf{v}$, so by §11.4 it is exactly what $\cos(\mathbf{u}, \mathbf{v})$ computes: $\cos(\mathbf{u}, \mathbf{v}) = \cos(\pi/4)$. Substituting,

$$\mathbf{w} \;=\; \frac{\mathbf{u}}{\|\mathbf{u}\|} \cdot \cos(\mathbf{u}, \mathbf{v}) \cdot \|\mathbf{v}\|.$$

Step 4 — unfold the cosine similarity. Now use the definition $\cos(\mathbf{u}, \mathbf{v}) = \langle \mathbf{u}, \mathbf{v}\rangle/(\|\mathbf{u}\|\|\mathbf{v}\|)$:

$$\mathbf{w} = \frac{\mathbf{u}}{\|\mathbf{u}\|} \cdot \frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|\,\|\mathbf{v}\|} \cdot \|\mathbf{v}\| = \frac{\mathbf{u}}{\|\mathbf{u}\|} \cdot \frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|} = \frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|^2}\,\mathbf{u}.$$

The $\|\mathbf{v}\|$ cancels, and with it the last trace of geometry: no angle, no protractor, no triangle. The shadow is computable from the components of $\mathbf{u}$ and $\mathbf{v}$ alone.

Written back in direction–magnitude form, the same vector reads

$$\mathbf{w} \;=\; \underbrace{\frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|}}_{\text{magnitude}} \;\cdot\; \underbrace{\frac{\mathbf{u}}{\|\mathbf{u}\|}}_{\text{direction}}\;.$$

12.4. Projection and component

The shadow vector has a formal name.

Definition 11 (Projection and component)**.** For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ with $\mathbf{u} \neq \mathbf{0}$, the **projection of v onto u** is

$$\operatorname{proj}_{\mathbf{u}}(\mathbf{v}) \;:=\; \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|} \cdot \frac{\mathbf{u}}{\|\mathbf{u}\|} \;=\; \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|^2}\, \mathbf{u},$$

and the scalar part of it — the length of the shadow — is the **component of v along u**,

$$\operatorname{comp}_{\mathbf{u}}(\mathbf{v}) \;:=\; \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|}.$$

The two are related by exactly the decomposition of §12.2 — component is the magnitude, $\hat{\mathbf{u}}$ is the direction:

$$\operatorname{proj}_{\mathbf{u}}(\mathbf{v}) \;=\; \operatorname{comp}_{\mathbf{u}}(\mathbf{v}) \cdot \hat{\mathbf{u}}.$$

The projection is a *vector*; the component is a *scalar*. Keeping that straight is most of the battle.

Remark (The component is signed). Although we called α "the length of the shadow," $\operatorname{comp}_{\mathbf{u}}(\mathbf{v})$ can be negative, because $\langle \mathbf{u}, \mathbf{v} \rangle$ can be. Reading it off the table in §11.4: the component is positive when θ is acute (so $\mathbf{v}$ leans toward $\mathbf{u}$), zero when the two are orthogonal (no shadow at all), and negative when θ is obtuse (so $\mathbf{v}$ leans away, and the shadow falls on the far side of the origin, pointing against $\mathbf{u}$). It is a *signed* length.

Remark (Order matters). The subscript names the vector you are projecting *onto*, and swapping the two arguments gives a different answer: $\operatorname{proj}_{\mathbf{u}}(\mathbf{v}) \neq \operatorname{proj}_{\mathbf{v}}(\mathbf{u})$ in general. The numerator $\langle \mathbf{u}, \mathbf{v} \rangle$ is symmetric, but the $\|\mathbf{u}\|^2$ in the denominator and the $\mathbf{u}$ outside it are not. Problems P1 and P2 below are the same two vectors in the two orders; compare the answers.

Example 21 (The motivating case, both ways)**.** Take $\mathbf{u} = [\,2,\ 0\,]$ along the x-axis and $\mathbf{v} = [\,3,\ 3\,]$, which does make an angle of $\pi/4$ with it. By the formula: $\langle \mathbf{u}, \mathbf{v} \rangle = 2(3) + 0(3) = 6$ and $\|\mathbf{u}\|^2 = 4$, so

$$\operatorname{comp}_{\mathbf{u}}(\mathbf{v}) = \frac{6}{2} = 3, \qquad \operatorname{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{6}{4}[\,2,\ 0\,] = \tfrac{3}{2}[\,2,\ 0\,] = [\,3,\ 0\,].$$

Now check it against the trigonometry of Step 2, which never mentioned components: $\|\mathbf{v}\| = \sqrt{18} = 3\sqrt{2}$ and $\cos(\pi/4) = \sqrt{2}/2$, so

$$\alpha = \|\mathbf{v}\| \cos\frac{\pi}{4} = 3\sqrt{2} \cdot \frac{\sqrt{2}}{2} = \frac{3 \cdot 2}{2} = 3.$$

The two routes agree. And the answer is the one the picture demands: the shadow of $[3, 3]$ on the x-axis reaches out to $x = 3$.

12.5. Practice: projections

For each pair, compute $\operatorname{comp}_{\mathbf{u}}(\mathbf{v})$ and $\operatorname{proj}_{\mathbf{u}}(\mathbf{v})$.

P1. $\mathbf{u} = [\,1,\ 0\,], \quad \mathbf{v} = [\,3,\ 4\,]$.

P2. $\mathbf{u} = [\,3,\ 4\,], \quad \mathbf{v} = [\,1,\ 0\,]$.

P3. $\mathbf{u} = [\,1,\ 2\,], \quad \mathbf{v} = [\,4,\ -3\,]$.

Answer key

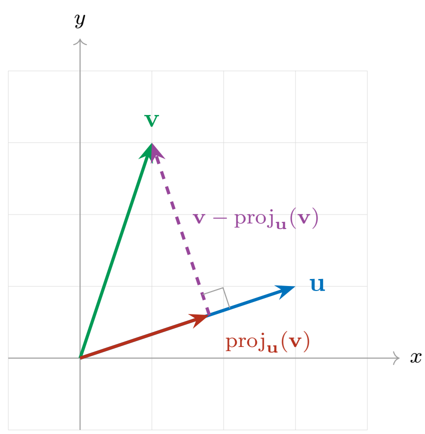
#	$\langle \mathbf{u}, \mathbf{v} \rangle$	$\lVert\mathbf{u}\rVert^2$	$\operatorname{comp}_{\mathbf{u}}(\mathbf{v})$	$\operatorname{proj}_{\mathbf{u}}(\mathbf{v})$
P1	$1(3) + 0(4) = 3$	1	$\dfrac{3}{1} = 3$	$\dfrac{3}{1}[1, 0] = [\,3,\ 0\,]$
P2	$3(1) + 4(0) = 3$	25	$\dfrac{3}{5}$	$\dfrac{3}{25}[3, 4] = \left[\,\frac{9}{25},\ \frac{12}{25}\,\right]$
P3	$1(4) + 2(-3) = -2$	5	$-\dfrac{2}{\sqrt{5}}$	$-\dfrac{2}{5}[1, 2] = \left[\,-\frac{2}{5},\ -\frac{4}{5}\,\right]$

Three things to take from these. **P1**: projecting onto $[\,1,\ 0\,]$ simply keeps the first coordinate and discards the second, which is what projecting onto the x-axis ought to do — a good sanity check that the formula is not doing anything exotic. **P2**: the same two vectors in the opposite order give a completely different answer, even though the dot product in the numerator is identical. **P3**: the component is negative, so the projection points *against* $\mathbf{u}$; here $\mathbf{v}$ leans away from $\mathbf{u}$, at an obtuse angle to it.

12.6. The orthogonal piece

Return to the picture. We dropped a perpendicular from the head of $\mathbf{v}$ down to $\mathbf{u}$ and used its foot to locate the shadow. But that perpendicular segment is itself a vector, and it is worth naming: it runs from the head of $\operatorname{proj}_{\mathbf{u}}(\mathbf{v})$ to the head of $\mathbf{v}$, so by the between-the-heads reading of §7.3 it is

$$\mathbf{v} - \operatorname{proj}_{\mathbf{u}}(\mathbf{v}).$$



Notice first that the two pieces reassemble $\mathbf{v}$, head to tail:

$$\mathbf{v} \;=\; \underbrace{\operatorname{proj}_{\mathbf{u}}(\mathbf{v})}_{\text{the part along } \mathbf{u}} \;+\; \underbrace{\big(\mathbf{v} - \operatorname{proj}_{\mathbf{u}}(\mathbf{v})\big)}_{\text{the part orthogonal to } \mathbf{u}}.$$

Subtracting off $\operatorname{proj}_{\mathbf{u}}(\mathbf{v})$ removes from $\mathbf{v}$ exactly its directional contribution along $\mathbf{u}$ — everything that pointed that way is gone, and what remains cannot point along $\mathbf{u}$ at all. The claim implicit in the labelling is that the remainder is genuinely orthogonal to $\mathbf{u}$, and it is short work to check.

Verification

Write $c = \langle \mathbf{u}, \mathbf{v} \rangle / \|\mathbf{u}\|^2$, so that $\mathrm{proj}_{\mathbf{u}}(\mathbf{v}) = c\,\mathbf{u}$ and $\mathbf{v} - \mathrm{proj}_{\mathbf{u}}(\mathbf{v}) = [\, v_1 - cu_1,\ v_2 - cu_2 \,]$. Dot it against $\mathbf{u}$:

$$\begin{aligned} \big\langle \mathbf{u},\ \mathbf{v} - \mathrm{proj}_{\mathbf{u}}(\mathbf{v}) \big\rangle &= u_1(v_1 - cu_1) + u_2(v_2 - cu_2) \\ &= \big(u_1 v_1 + u_2 v_2\big) - c\big(u_1^2 + u_2^2\big) \\ &= \langle \mathbf{u}, \mathbf{v} \rangle - c\,\|\mathbf{u}\|^2 \\ &= \langle \mathbf{u}, \mathbf{v} \rangle - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|^2} \cdot \|\mathbf{u}\|^2 \ = \ 0. \end{aligned}$$

By the orthogonality test of §11.7, a dot product of zero means orthogonal. Note which identity did the work in the third line: $u_1^2 + u_2^2 = \|\mathbf{u}\|^2$, the $\langle \mathbf{u}, \mathbf{u} \rangle = \|\mathbf{u}\|^2$ of §11.6. That is the promised payoff — the identity is what lets $\|\mathbf{u}\|^2$ cancel against the denominator of c.

Looking ahead. Splitting a vector into a piece along $\mathbf{u}$ and a piece orthogonal to $\mathbf{u}$ is not a one-off trick; it is the engine of a great deal of computational linear algebra. Run it repeatedly on a list of vectors and you manufacture a whole orthogonal family out of an arbitrary one — which is the idea behind the Gram–Schmidt procedure and, through it, the QR factorization and the numerical algorithms built on top of them. We will meet all of these later; for now, notice only how little machinery it took: a dot product, a norm, and a subtraction.

12.7. The Cauchy–Schwarz inequality

The projection picture has one more thing to tell us, and it is a bound on the dot product itself.

Look again at the right triangle of §12.3. Its hypotenuse is $\mathbf{v}$, with length $\|\mathbf{v}\|$, and one of its legs is the shadow, with length $|\mathrm{comp}_{\mathbf{u}}(\mathbf{v})|$. A leg of a right triangle is never longer than the hypotenuse — equivalently, and more memorably:

> A shadow is never longer than the object casting it.

In symbols, $|\mathrm{comp}_{\mathbf{u}}(\mathbf{v})| \ \le\ \|\mathbf{v}\|$. Now substitute the definition of the component and clear the denominator:

$$\left| \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|} \right| \ \le\ \|\mathbf{v}\| \qquad \Longrightarrow \qquad |\langle \mathbf{u}, \mathbf{v} \rangle| \ \le\ \|\mathbf{u}\|\,\|\mathbf{v}\|.$$

That is a bound on how large a dot product can possibly be, in terms of nothing but the two magnitudes — and it is famous enough to carry two names.

> **Cauchy–Schwarz inequality.** For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$,
>
> $$\big|\langle \mathbf{u}, \mathbf{v} \rangle\big| \ \le\ \|\mathbf{u}\|\,\|\mathbf{v}\|.$$

There is also a one-line route to it from §11 that requires no picture at all. Since $\langle \mathbf{u}, \mathbf{v} \rangle = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta$ and $|\cos\theta| \le 1$ for every angle,

$$\big|\langle \mathbf{u}, \mathbf{v} \rangle\big| = \|\mathbf{u}\|\,\|\mathbf{v}\|\,|\cos\theta| \le \|\mathbf{u}\|\,\|\mathbf{v}\|.$$

This second route also settles when the bound is *tight*: equality needs $|\cos\theta| = 1$, hence $\theta = 0$ or $\theta = \pi$. So the dot product attains its largest possible size exactly when the two vectors are parallel or antiparallel — when, in the shadow picture, the triangle has collapsed and $\mathbf{v}$ lies flat along $\mathbf{u}$, casting a shadow as long as itself.

Example 22 (The bound in action)**.** For $\mathbf{u} = [\,3,\,1\,]$ and $\mathbf{v} = [\,1,\,3\,]$ we found $\langle \mathbf{u}, \mathbf{v}\rangle = 6$ in §11, while $\|\mathbf{u}\|\|\mathbf{v}\| = \sqrt{10} \cdot \sqrt{10} = 10$. And $6 \le 10$, with room to spare — the slack is exactly the factor $\cos\theta = 0.6$. For $\mathbf{u} = [\,1,\,2\,]$ and $\mathbf{v} = [\,2,\,4\,]$, which are parallel, we get $\langle \mathbf{u}, \mathbf{v}\rangle = 10$ and $\|\mathbf{u}\|\|\mathbf{v}\| = \sqrt{5} \cdot 2\sqrt{5} = 10$: equality, as predicted.

Remark (Why it matters more than it looks)*.* Cauchy–Schwarz is what makes §11.4 legitimate. We defined $\cos(\mathbf{u}, \mathbf{v}) = \langle \mathbf{u}, \mathbf{v}\rangle / (\|\mathbf{u}\|\|\mathbf{v}\|)$ and asserted that the output lands in $[-1, 1]$ — but rearrange the inequality and that assertion *is* Cauchy–Schwarz:

$$\big|\langle \mathbf{u}, \mathbf{v}\rangle\big| \le \|\mathbf{u}\|\|\mathbf{v}\| \qquad \Longleftrightarrow \qquad -1 \;\le\; \frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|\,\|\mathbf{v}\|} \;\le\; 1.$$

In $\mathbb{R}^2$ we could lean on the angle θ, which we can see. In higher dimensions, and in the abstract inner-product spaces of later chapters, there is no picture to look at and the angle is *defined* by that quotient — so something has to guarantee the quotient is a legal cosine before arccos may be applied to it. Cauchy–Schwarz is that guarantee, which is why it is stated as an inequality about inner products rather than as a fact about triangles.

Remark (A second dividend: the triangle inequality, again)*.* In §7.2 we accepted $\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\|+\|\mathbf{v}\|$ because a triangle cannot have one side longer than the other two combined — true, but geometric. It now follows algebraically. Expanding as in §11.2,

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v}\rangle + \|\mathbf{v}\|^2 \;\le\; \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2 = \big(\|\mathbf{u}\| + \|\mathbf{v}\|\big)^2,$$

where the middle step is Cauchy–Schwarz. Taking square roots of both non-negative sides gives the triangle inequality. The geometric picture and the algebraic bound were the same statement all along.

13. Practice set: addition and subtraction

For each pair below: **(a)** draw $\mathbf{u}$ and $\mathbf{v}$ on one set of axes, tails at the origin, arrowheads marked; **(b)** compute the indicated sum or difference componentwise; **(c)** draw the resulting vector on the same axes.

P1. $\mathbf{u} = [3, 1]$, $\quad \mathbf{v} = [1, 2]$. $\quad$ Compute $\mathbf{u} + \mathbf{v}$.

P2. $\mathbf{u} = [4, 1]$, $\quad \mathbf{v} = [1, 3]$. $\quad$ Compute $\mathbf{u} - \mathbf{v}$.

P3. $\mathbf{u} = [-2, 3]$, $\quad \mathbf{v} = [3, 1]$. $\quad$ Compute $\mathbf{u} + \mathbf{v}$.

P4. $\mathbf{u} = [2, -2]$, $\quad \mathbf{v} = [-3, -1]$. $\quad$ Compute $\mathbf{u} - \mathbf{v}$.

Optional: compute the norm of each resultant.

Watch the signs in P4: both inputs have a negative component, so $2-(-3) = 5$ and $-2-(-1) = -1$.

Answer key

#	$\mathbf{u}$	$\mathbf{v}$	**Result**	$\lVert\text{result}\rVert$
P1	$[3, 1]$	$[1, 2]$	$\mathbf{u} + \mathbf{v} = [4, 3]$	5
P2	$[4, 1]$	$[1, 3]$	$\mathbf{u} - \mathbf{v} = [3, -2]$	$\sqrt{13}$
P3	$[-2, 3]$	$[3, 1]$	$\mathbf{u} + \mathbf{v} = [1, 4]$	$\sqrt{17}$
P4	$[2, -2]$	$[-3, -1]$	$\mathbf{u} - \mathbf{v} = [5, -1]$	$\sqrt{26}$